

SSI-DDI: substructure–substructure interactions for drug–drug interaction prediction

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Abstract

A major concern with co-administration of different drugs is the high risk of interference between their mechanisms of action, known as adverse drug–drug interactions (DDIs), which can cause serious injuries to the organism. Although several computational methods have been proposed for identifying potential adverse DDIs, there is still room for improvement. Existing methods are not explicitly based on the knowledge that DDIs are fundamentally caused by chemical substructure interactions instead of whole drugs' chemical structures. Furthermore, most of existing methods rely on manually engineered molecular representation, which is limited by the domain expert's knowledge. We propose substructure–substructure interaction–drug–drug interaction (SSI-DDI), a deep learning framework, which operates directly on the raw molecular graph representations of drugs for richer feature extraction; and, most importantly, breaks the DDI prediction task between two drugs down to identifying pairwise interactions between their respective substructures. SSI-DDI is evaluated on real-world data and improves DDI prediction performance compared to state-of-the-art methods. Source code is freely available at <https://github.com/kanz76/SSI-DDI>.

Key words: drug–drug interactions; substructure interactions; molecular graph; co-attention; multi-type interactions

Introduction

Even though co-administration of two or more drugs is common practice when dealing with patients with multiple illnesses [1] or when the synergy of multiple medications is needed to treat diseases with complex biological processes [2, 3], the drugs under consideration do not always interact with each other as expected. They can result in side effects capable of putting a patient's life at risk. These side effects are caused by the chemical-physical interactions, usually termed drug–drug interactions (DDIs), of the drugs taken together. It has been estimated that DDIs are associated with 30% of all the reported adverse side effects and are one of the major reasons for drug withdrawal from the market [4], resulting both in a large number of morbidity and mortality [1], and also incurring huge medical costs due to the immense cost of drug design [5]. DDIs have, therefore,

become the focus of research and clinical studies. Many research works are being conducted, ranging from DDI extraction [6] to DDI prediction tasks. The present work belongs to the latter category, that is, it is a (computational) research study on the prediction of DDIs between a pair of drugs.

Systematic screening of DDI candidates during clinical testing can be limited by manpower and fails to handle the fast growing amount of biomedical data [7]. This leaves many drugs on the market with unknown adverse interactions with each other. Automated computational approaches (such as machine and deep learning methods) have proved to be essential for ameliorating this problem.

Various machine learning methods [8–16] have been proposed and have greatly contributed to the DDI prediction task. The vast majority of these methods rely on the drug similarity

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assumption, where it is believed that if drugs A and B interact to produce a specific biological effect, then drugs similar to drug A (or drug B) are likely to interact with drug B (or drug A) to produce the same effect [17]. Drugs are, therefore, processed depending on their similarities in chemical structures; and/or additionally, in other features such as their individual side effects, targets, pathways, etc. For instance, Zhang et al. [15] build an ensemble of prediction methods based on 14 kinds of drug-drug similarities from chemical, biological, phenotypic and topological data. Zhang et al. [16] propose a matrix completion method for DDI prediction where 8 kinds of drug-drug similarities are used as part of regularization for conserving drugs' manifold in lower space. The downside with this assumption is that similar drugs (or chemical entities, in general) do not necessarily have the same biological activity [5]. The features for which they show some similarity might be irrelevant to the prediction task of concern. Furthermore, methods that consider additional features (such as sideeffects, pathways) are often constrained to work with smaller datasets, given that these features are not always available for all the drugs. Some features such as individual drug sideeffects can only be available post-market in a real-world scenario, making these methods unsuitable for application in early drug development phase.

Recently, many deep learning methods [4, 18–26] have been developed and are improving even further DDI prediction tasks, showing encouraging performance. For instance, Zhang et al. [24] propose a multi-modal deep auto-encoders to work with incomplete, large-scale and noisy drug data. Feng et al. [25] considers DDIs as a network on which a deep graph auto-encoder is applied for latent representations of drugs, which are then fed to a deep feedforward neural network for DDI prediction. However, even though the learning capability of deep learning have shown to be effective, it is still mostly unexplored in DDI prediction tasks, especially as far as feature extraction from the raw representations, i.e. chemical structures, of drugs are concerned. Some of the existing methods still rely on the similarity assumption of drugs or on manually engineered features (e.g. molecular fingerprints [27]). The latter require expert domain knowledge, which can be labor intensive, and may contain noisy information for downstream tasks. Moreover, these features are designed in a way that they capture the whole molecular structure of drugs, while it has been proven that DDIs usually depend only on a few substructures of the whole chemical structures [2, 5]. Considering the whole molecular structure could bias the prediction toward irrelevant data. MR-GNN [19] and GoGNN [21] are two recent methods that both take the aforementioned issues into account: (i) They both leverage the powerful feature extraction ability of deep learning by working directly on raw molecular graph representations of chemical structures of drugs using graph neural networks (GNNs) [28–31]. (2) They both consider, rather in an implicit manner, the important role of substructures in DDI prediction: MR-GNN uses LSTMs [32], artificial recurrent neural networks, to learn a comprehensive feature representation of a drug from different-size substructures, and GoGNN represents a drug as a concatenation of its different substructures. In this paper, we propose a deep learning-based method (Figure 1), SSI-DDI, which similarly leverages GNNs for powerful feature extraction. However, compared to these methods, our contributions are:

- A knowledge-driven model that explicitly consider substructures as independent entities whose pairwise

interactions will determine whether the drugs from which they are extracted can result in a potential DDI.

- A deep learning model that learns those substructures directly from the molecular graphs of the drugs. And because these pairwise substructure interactions will not contribute equally to the final DDI prediction, the model learns as well the relevance/importance of each one of them to the final DDI prediction using a co-attention mechanism [33] (similar to interaction map [34–36]). As a by-product, this can improve the ability of both expert and non-expert users to interpret the results of the prediction.

We show that our work outperforms previous methods and can also be used in an inductive setting (also known as cold-start scenario), where DDI prediction can be made on drugs that the model was not trained on.

Method

This section gives a formal definition of the problem we are trying to solve, and presents the building blocks of our framework, including the input representation and the computation steps, in a sequential order. The overall framework is shown in Figure 1.

Problem settings

Given a set of drugs $\mathcal{G}$, a set of interaction types $\mathcal{I} = \{I_i\}_{i=1}^M$, and a dataset of DDIs $\mathcal{D} = \{(G_x, G_y, r)_{i=1}^N\}$, where $G_x \in \mathcal{G}$ and $G_y \in \mathcal{G}$ represent a pair of drugs with interaction r of type I_i , we want to find a model $f : \mathcal{G} \times \mathcal{G} \times \mathcal{I} \rightarrow [0, 1]$ which can determine the probability that a combination of any two drugs can result in a given interaction type I_i .

Inputs

The model takes as input a DDI tuple (G_x, G_y, r) . Drugs G_x and G_y are both represented as graphs $G = (V, E)$, where $V = \{v_i\}_{i=1}^n$ is the set of nodes and $E = \{(v_s, v_t)\}_{i=1}^m$ the set of edges; and each interaction type $I_r \in \mathcal{I}$ is represented as a learnable matrix $\mathbf{M}_r \in \mathbb{R}^{D \times D}$. The interpretation of each node v_i will depend on how many convolution operations have been applied to it (see Section Chemical substructure extraction). Initially, $G = (V, E)$ simply represents the molecular graph of the chemical structure of a drug, where v_i represents an atom with a feature vector $\mathbf{h}_i \in \mathbb{R}^F$ (F is the number of features), and (v_s, v_t) indicates that there is a bond between atoms v_s and v_t .

Graph convolution operator

Duvenaud et al. [37] and Kearnes et al. [38] show that features learned directly from the molecular graphs of chemical entities in an end-to-end approach are much more informative for downstream tasks than handcrafted molecular representations such as molecular fingerprints. They particularly use GNNs, a deep learning method specifically designed for graph-structured data. GNNs are fundamentally composed of convolution operators/layers where a node's features are updated with the aggregation of its surrounding/adjacent nodes' features just as a pixel's channels update is performed in the field of computer vision using convolutional neural networks [39]. In this work, we use L GAT layers [28] as convolution operators for node features

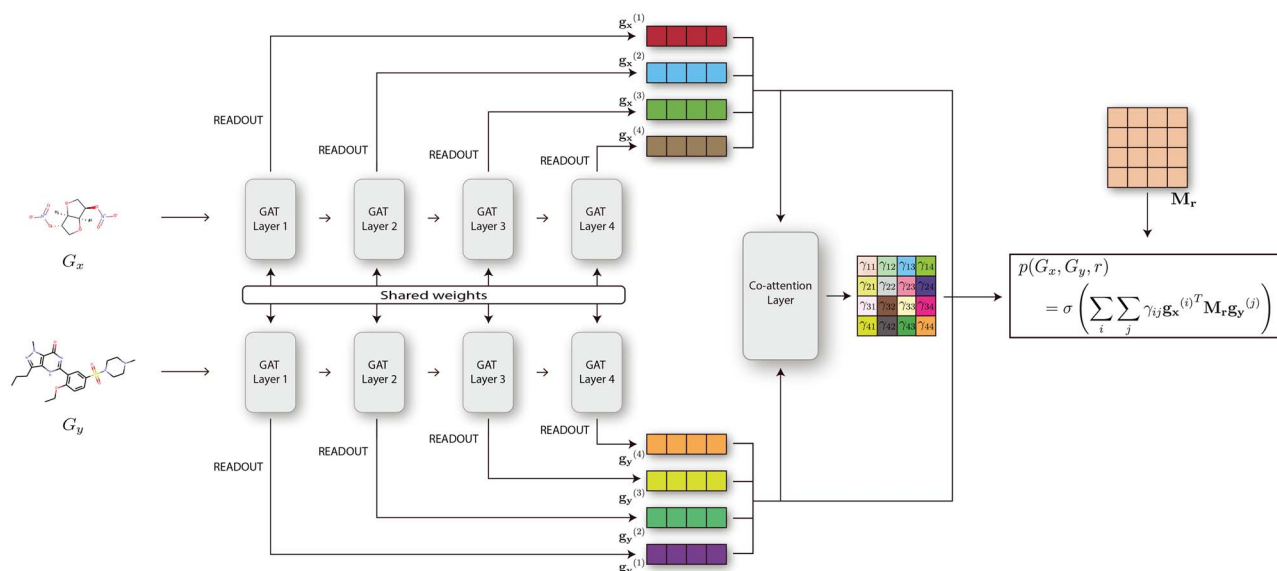


Figure 1. Overview of substructure–substructure interaction–drug–drug interaction (SSI-DDI) workflow with L graph attention (GAT) layers (here, $L = 4$ to keep the figure uncluttered). To determine if two drugs, G_x and G_y , can result in interaction r (represented as a matrix M_r): (1) The model takes as input the graphical representations of the pair of drugs (Sections Problem settings and Inputs); (2) At each GAT layer, a READOUT operation is performed to extract the substructure information (e.g. $g_x^{(1)}$, $g_y^{(3)}$) within the reach of the receptive field of the layer (Section Chemical substructure extraction); (3) A co-attention layer measures the relevance (to the final DDI prediction) of each interaction score/intensity between substructures of G_x and G_y using a co-attention mechanism (e.g. γ_{13} is the relevance of the interaction between $g_x^{(1)}$ and $g_y^{(3)}$) (Section Substructure interaction relevance with co-attention); and (4) The final DDI prediction $p(G_x, G_y, r)$ is the sum of interaction scores ($g_x^{(i)T} M_r g_y^{(j)}$) of these substructures, each one weighted by its relevance γ_{ij} (Section Prediction and loss function).

update. In order to update node i 's feature vector $h_i^{(l+1)}$ at the $(l+1)$ th GAT layer, the latter:

- applies a linear transformation to the features of the previous l th GAT layer of all the nodes in the graph

$$\hat{h}_s^{(l+1)} = W^{(l+1)} h_s^{(l)} \quad s = 1, \dots, n, \quad (1)$$

where $W^{(l+1)} \in \mathbb{R}^{F \times F}$ is the learnable transformation matrix from the l th GAT layer to the $(l+1)$ th GAT layer.

- determines the importance $\alpha_{ij} \in \mathbb{R}$ of each (adjacent) node j in the neighborhood $\mathcal{N}_{(i)}$ of node i using the attention mechanism [40, 41]. This importance accounts for the fact that not all of the nodes $j \in \mathcal{N}_{(i)}$ will be equally relevant in updating node i 's feature vector, a learnable importance weight α_{ij} is therefore assigned to each one of them. Self-importance α_{ii} of node i is also considered in order to retain some information from itself. The importance α_{ij} of node j to node i is computed as follows:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(e_{ij}))}{\sum_{s \in \mathcal{N}_{(i)} \cup \{i\}} \exp(\text{LeakyReLU}(e_{is}))}, \quad (2)$$

where LeakyReLU [42] is an activation function (which mitigates the zero gradient issue with rectified linear unit (ReLU) activation function), and e_{ij} is the dot product of a learnable weight vector $a \in \mathbb{R}^{2F}$ with the concatenation of $\hat{h}_i^{(l+1)}$ and $\hat{h}_j^{(l+1)}$ Equation (1).

$$e_{ij} = a^T [\hat{h}_i^{(l+1)} \parallel \hat{h}_j^{(l+1)}] \quad (3)$$

T is the transposition operator that can also perform a dot product operation between two vectors, and $\parallel$ denotes concatenation.

- updates node i by aggregating all the nodes in $\mathcal{N}_{(i)} \cup \{i\}$, each one scaled by its importance α_{ij} Equation (2). This produces the pre-activation feature vector:

$$z_i^{(l+1)} = \sum_{s \in \mathcal{N}_{(i)} \cup \{i\}} \alpha_{is} \hat{h}_s^{(l+1)} \quad (4)$$

In order to use this feature vector as input to the next GAT layer, if any, an activation function is applied to capture complex non-linear signals, producing the activated feature vector of node i as follows:

$$h_i^{(l+1)} = \sigma(z_i^{(l+1)}), \quad (5)$$

where σ is an activation function. In our work, we use the exponential linear unit (ELU) [43] activation function, similar to [28]; which also, empirically, performed better than the most popular ReLU activation function.

Multi-head Attention

To stabilize the learning process and have information from different representation subspaces at different positions [44], we also employ multi-head attention mechanism with K heads where the GAT layer applies K linear transformations in parallel to every node s using Equation (1) to obtain $[\hat{h}_s^{(k)}]^{(l+1)} = [W^{(k)}]^{(l+1)} h_s^{(l)}$ for the k th head ($k = 1, \dots, K$). From Equations (2)–(4), the k th head produces $[z_i^{(k)}]^{(l+1)}$ for node i whose activated representation $h_i^{(l+1)}$ is obtained after the concatenation of the outputs of these K heads:

$$h_i^{(l+1)} = \sigma\left(\parallel_{k=1}^K [z_i^{(k)}]^{(l+1)}\right), \quad (6)$$

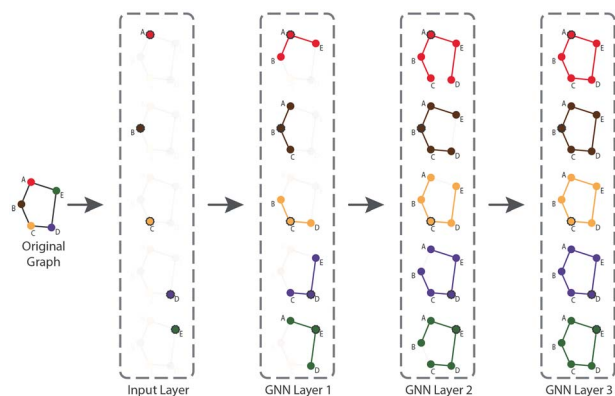


Figure 2. Example of substructure extraction in a three-layer GNNs adapted from [19]. In the original graph, each node, corresponding to an atom, is represented with a different color. At each GNN layer, a node (surrounded by a black ‘zig-zag’ ring) is updated by aggregating all its surrounding nodes within the reach of the receptive field of that layer. These surrounding nodes are shown turned into the color of the node being updated. After node update in a GNN layer, ‘each node’ can therefore represent ‘a substructure’, composed of its surrounding nodes, of which it is the ‘center’.

where $\parallel$ denotes concatenation.

Normalization

In order to increase converge speed and stabilize the learning process as well, we add a normalization layer after the output of each GAT layer, before applying the activation function σ . We use LayerNorm layers [45] which, empirically, performed the best. Taking normalization into account, Equation (5) becomes:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\text{LayerNorm} \left(\mathbf{z}_i^{(l+1)} \right) \right) \quad (7)$$

Equation (6) can be modified in the same way. We also apply an initial normalization to the input data before being fed to the model as a normalization preprocessing.

Initially, node i ’s features are simply the features of the atom it is associated with. With the effect of the convolution operation, node i ’s features start incorporating features of nodes to which it is linked to. Therefore, node i no longer represents a simple atom, but a subgraph/substructure comprising itself and its surrounding nodes/atoms. See Section Chemical substructure extraction and Figure 2 for more details.

Chemical substructure extraction

Carefully investigated, convolution operations gather the information of different substructures, each centered at a node i (Figure 2)—that is, the updated feature vector $\mathbf{h}_i^{(l+1)}$ [Equation (5)] (similarly, its counterpart pre-activation vector $\mathbf{z}_i^{(l+1)}$) contains the information of the substructures formed by the nodes $s \in \mathcal{N}_{(i)} \cup \{i\}$. The size of the substructures grows as we move from one layer to the next one. Additionally, the attention mechanism Equation (2) will make it possible that not all the nodes in a substructure are considered equally, refining even further the type of substructures being learned.

At each GAT layer l , information from all the substructures [represented by individual node feature vectors $\mathbf{z}_i^{(l)}$ Equation (4)] are aggregated (‘READOUT’ operation), each one weighted by a learnable coefficient $\beta_i \in \mathbb{R}$, which can be interpreted as its importance. Here, we use $\mathbf{z}_i^{(l)}$ instead of $\mathbf{h}_i^{(l)}$ because we

simply need the output of the current layer l rather than the non-linearly modified output $\mathbf{h}_i^{(l)}$, which is usually needed to be fed to the next layer. The representation of the drug $G_x \in \mathcal{G}$ capturing the information of all substructures identified at layer l is therefore given by the vector:

$$\mathbf{g}_x^{(l)} = \sum_{i=1}^n \beta_i \mathbf{z}_i^{(l)}. \quad (8)$$

We use SAGPooling [46] to determine the importance β_i of every node in the graph. Given a graph with feature matrix $\mathbf{X}$ and adjacency matrix $\mathbf{A}$, SAGPooling computes the importance vector β (whose components are the coefficients β_i) as follows:

$$\beta = \text{softmax}(\text{GNN}(\mathbf{A}, \mathbf{X})) \quad (9)$$

SAGPooling considers both the contextual and topological information of nodes to determine how important they are in the whole graph.

Substructure interaction relevance with co-attention

After obtaining all the substructure information $\mathbf{g}_x^{(l)}$ and $\mathbf{g}_y^{(l)}$ of the input drugs G_x and G_y , respectively (Section Chemical substructure extraction), from all the GAT layers, we use a co-attention mechanism [33] to account for the importance γ_{ij} of each pairwise interaction between the substructures of G_x and G_y which is given by:

$$\gamma_{ij} = \mathbf{b}^T \tanh \left(\mathbf{W}_x \mathbf{g}_x^{(i)} + \mathbf{W}_y \mathbf{g}_y^{(j)} \right) \quad (10)$$

$$i = j = 1, \dots, L,$$

where $\mathbf{b}$ is a learnable weight vector, $\mathbf{W}_x$ and $\mathbf{W}_y$ are learnable weight matrices. We use different weight matrices to avoid the situation where similar substructures are given high scores. This can complicate the whole learning process. As opposed to Equations (2) and (9), no activation function is used in Equation (10). We have to account for non-interacting drugs which we expect to have negative scores in order to produce very low DDI probability using Equation (11). Nonetheless, activation functions that output both positive and negative values such as $\tanh$ can be used, but we find that performance is better without any activation functions.

Prediction and loss function

The prediction of G_x and G_y having an interaction r of type I_r is given by the probability

$$P(G_x, G_y, r) = \sigma \left(\sum_i \sum_j \gamma_{ij} \mathbf{g}_x^{(i)T} \mathbf{M}_r \mathbf{g}_y^{(j)} \right) \quad (11)$$

where σ is the sigmoid function, and $\mathbf{M}_r$ the learnable matrix representing the interaction type I_r . Equation (11) considers the probability of a pair of drugs (G_x, G_y) to have an interaction r to be the aggregation of pairwise chemical reactions between the functional substructures of these drugs, each one of these combinations weighted by how much important or relevant it is to the final DDI prediction.

The model is trained end-to-end with gradient descent using binary cross-entropy. The given DDIs in the dataset $\mathcal{D}$ (Section Inputs) are considered as positive samples. For each positive sample $(G_x, G_y, r)_i$, a negative sample is generated by either replacing G_x or G_y following the strategy proposed in [47]. The loss function $\mathcal{L}$ on the entire DDI dataset $\mathcal{D}$ is computed as follows:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N (\log(p_i) + \log(-p'_i)), \quad (12)$$

where both p_i and p'_i are computed using Equation (11). p_i is used for a positive sample and p'_i is used for its associated negative sample.

SSI-DDI overall algorithm

The overall algorithm of our method is shown in Algorithm 1. A pair of drug molecular graphs G_x and G_y , and their interaction r are provided as inputs. Atom features are stacked as row vectors forming features matrices $\mathbf{H}_x$ and $\mathbf{H}_y$ and are used to initialize nodes in the graphical representations of G_x and G_y , respectively (Line 5). At each GAT layer (Lines 9–10), feature matrices are updated (Lines 11–12), and substructures are extracted (Lines 13–14). The relevance of a pairwise interaction is computed in Line 18. Line 22 determines the DDI probability.

Algorithm 1 SSI-DDI Overall Algorithm

```

1: Input:                                     // Section 2.2
2:   Molecular graph  $G_x$ , atom feature matrix  $\mathbf{H}_x$ 
3:   Molecular graph  $G_y$ , atom feature matrix  $\mathbf{H}_y$ 
4:   Interaction  $r$  of type  $I_r$ 
5: Initialize:
6:    $\mathbf{H}_x^{(0)} \leftarrow \mathbf{H}_x$ 
7:    $\mathbf{H}_y^{(0)} \leftarrow \mathbf{H}_y$ 
8: for  $l \leftarrow 1, L$  do
9:    $\mathbf{Z}_x^{(l)} \leftarrow \text{GAT}^{(l)}(G_x, \mathbf{H}_x^{(l-1)})$ 
10:   $\mathbf{Z}_y^{(l)} \leftarrow \text{GAT}^{(l)}(G_y, \mathbf{H}_y^{(l-1)})$     // Lines 9-10 Eq. (1-4)
11:   $\mathbf{H}_x^{(l)} \leftarrow \sigma(\mathbf{Z}_x^{(l)})$ 
12:   $\mathbf{H}_y^{(l)} \leftarrow \sigma(\mathbf{Z}_y^{(l)})$                 // Lines 11-12 Eq. (5)
13:   $\mathbf{g}_x^{(l)} \leftarrow \text{READOUT}(\mathbf{Z}_x^{(l)})$ 
14:   $\mathbf{g}_y^{(l)} \leftarrow \text{READOUT}(\mathbf{Z}_y^{(l)})$         // Lines 13-14 Eq. (9)
15: end for
16: for  $i \leftarrow 1, L$  do
17:   for  $j \leftarrow 1, L$  do
18:     $\gamma_{ij} \leftarrow \text{Co-attention}(\mathbf{g}_x^{(i)}, \mathbf{g}_y^{(j)})$     // Eq. (10)
19:   end for
20: end for
21:  $\mathbf{M}_r \leftarrow \text{getInteractionMatrix}(\text{type}=I_r)$ 
22:  $P(G_x, G_y, r) \leftarrow \sigma\left(\sum_i \sum_j \gamma_{ij} \mathbf{g}_x^{(i)T} \mathbf{M}_r \mathbf{g}_y^{(j)}\right)$     // Eq. (11)
23: return  $P(G_x, G_y, r)$ 

```

Experiments

Dataset

We used the DrugBank¹ dataset (version 5.0.3) as preprocessed and split (6:2:2 ratio) by [4]² for the DDI prediction task. The dataset contains 1704 drugs and 191 400 DDIs classified into 86 general sentence structures, each of which describing a specific DDI type.

Drugs are associated with their SMILES string representations [48], which were converted into graphical representations using the open-source cheminformatics library RDKit³. For each drug, a graph was generated where bonds were represented as edges, and atoms as nodes. For each atom v_i , the feature vector $\mathbf{h}_i \in \mathbb{R}^{55}$ contained:

- The atomic symbol (44, one-hot representation)
- The degree of the atom (number of adjacent atoms)
- The implicit valence
- The formal charge
- The number of radical electrons
- The hybridization (5, one-hot representation)
- Whether or not the atom is aromatic
- The total number of Hydrogens on the atom.

Setup

Our model, SSI-DDI, is composed of GAT layers with shared weights for learning the representations of a pair of drugs (Figure 1). This can simply be implemented by reusing the same layers for both drugs. Using this setting, our model consists of $L = 4$ GAT layers, each one with a multi-head attention of $K = 2$ heads. Each head computes 32 intermediate features, bringing the final output of a GAT layer to a hidden feature vector of dimension $64 (2 \times 32)$. A GAT layer is followed by a LayerNorm layer and ELU activation function. A LayerNorm layer is also applied directly to the input data as a normalization preprocessing operation. Each interaction type $I_r \in \mathcal{I}$ is represented by a learnable matrix $\mathbf{M}_r \in \mathbb{R}^{64 \times 64}$.

The model is trained on mini-batches of 1024 DDIs using the Adam optimizer [49] with weight decay set to $5e - 4$. We set a learning rate with an exponentially decaying schedule $lr = 1e - 2 \times 0.96^t$, where t is the current epoch.

Implementation

Our model is implemented in PyTorch Geometric [50]. All experiments were run on Windows 10 OS with a NVIDIA GeForce RTX 2070 GPU, a processor Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz and 32GB of RAM.

Baselines

We compare our method with the following baselines.

- **MR-GNN** [19]: a GNN that can learn comprehensive and effective features by leveraging a multi-resolution architecture.
- **GoGNN** [21]: a model that leverages the dual attention mechanism in the view of graph of graphs to capture the information from both entity graphs and entity interaction graph hierarchically.
- **MHCADDI** [26]: a GNN architecture that introduces external message passing mechanism between drugs' structures to integrate joint drug-drug information during the representation learning phase for individual drugs.
- **DeepDDI** [4]: one of the early adopters of deep learning for DDI prediction. DeepDDI is a similarity-based method that designs a feature structural similarity profiles for each drug based on their molecular fingerprints. Pairs of drugs' similarity profile are then fed to a multi-layer perceptron for prediction.

¹ <https://go.drugbank.com/releases>

² <https://www.pnas.org/content/suppl/2018/04/14/1803294115>. DC Supplemental)

³ <https://www.rdkit.org>

Table 1. Comparative evaluation on the DrugBank dataset

	ACC	AUROC	AUPRC
DeepDDI	0.9315	0.9976	0.9319
MR-GNN	0.9406	0.9976	0.9410
GoGNN	0.8478	0.9163	0.9001
MHCADDI	0.7850	0.8633	0.8398
SSI-DDI(Ours)	0.9447	0.9838	0.9814

MR-GNN and DeepDDI treat the DDI task as a multi-class classification problem where the input is a pair of drugs and the output is the most probable interaction type that may occur from this combination of drugs. GoGNN and MHCADDI, on the other hand, similar to our method, find the probability of a triple of two drugs and a given interaction type.

Results

We evaluate the experimental results using three metrics: the ‘accuracy’ (ACC), the ‘area under the receiver operating characteristic’ (AUROC) and the ‘area under the precision recall curve’ (AUPRC). For multi-class classification models (i.e. MR-GNN and DeepDDI), we consider the micro metric of AUROC and AUPRC. Each method was run on the same split of the dataset (See Section Dataset). For MR-GNN, we reused the original implementation made available by the authors. Go-GNN was slightly modified by changing the way the metrics are computed: in the original work, prediction is made by comparing the score of a positive sample with the one of its associated negative sample; in the modified version both scores are considered independently, as it is the case with the other methods. We re-implemented the DeepDDI and MHCADDI methods following the settings given in their respective original papers. Table 1 shows the performance of the different methods on the test set in transductive setting. Our method outperforms the others in ACC and AUPRC (with a significant margin of 4.04%). Even though MR-GNN and DeepDDI show high performance in AUROC, this metric is not very reliable due to the imbalance nature of the DrugBank dataset (see [Supplementary materials](#), section 1). Multi-class classifiers tends to have very high AUROC scores, even if the true positive rate is very low. Our method, however, has both very high AUROC and AUPRC scores, showing that not only it distinguishes between interacting and non-interacting drugs, but does it with high precision.

Inductive setting

So far, we presented the results of experiments in a transductive setting, that is, the drugs in test phase were also included in the training phase. Next, in order to evaluate our method’s performance when presented with new drugs on which it was not trained on (also known as cold start problem), we split the dataset with respect to the drugs instead of DDIs. Let $\mathcal{G}_{new}$ denotes the set of new drugs (unseen during training), and $\mathcal{G}_{old}$ denotes the seen drugs, such that $\mathcal{G}_{new} \cup \mathcal{G}_{old} = \mathcal{G}$ and $\mathcal{G}_{new} \cap \mathcal{G}_{old} = \emptyset$, the new dataset split consists of:

$$\begin{aligned}\mathcal{D}_{train} &= \{(G_x, G_y, r) \in \mathcal{D} \mid G_x \in \mathcal{G}_{old} \wedge G_y \in \mathcal{G}_{old}\} \\ \mathcal{D}_{S1} &= \{(G_x, G_y, r) \in \mathcal{D} \mid G_x \in \mathcal{G}_{new} \wedge G_y \in \mathcal{G}_{new}\} \\ \mathcal{D}_{S2} &= \{(G_x, G_y, r) \in \mathcal{D} \mid (G_x \in \mathcal{G}_{old} \wedge G_y \in \mathcal{G}_{new}) \\ &\quad \vee (G_x \in \mathcal{G}_{new} \wedge G_y \in \mathcal{G}_{old})\}\end{aligned}$$

Table 2. Comparative evaluation (mean $\pm$ SD) in inductive setting (%)

	ACC	AUROC	AUPRC
$\mathcal{D}_{S1}$ partition			
DeepDDI	47.96 $\pm$ 2.66	88.39 $\pm$ 2.19	48.27 $\pm$ 2.64
MR-GNN	49.93 $\pm$ 2.32	91.70 $\pm$ 0.90	50.23 $\pm$ 2.30
MHCADDI	66.53 $\pm$ 1.05	72.13 $\pm$ 1.24	72.48 $\pm$ 1.12
SSI-DDI(Ours)	65.02 $\pm$ 0.74	72.38 $\pm$ 1.01	72.83 $\pm$ 1.35
$\mathcal{D}_{S2}$ partition			
DeepDDI	68.69 $\pm$ 1.76	95.75 $\pm$ 1.12	68.87 $\pm$ 1.75
MR-GNN	72.38 $\pm$ 0.73	97.52 $\pm$ 0.31	72.54 $\pm$ 0.73
MHCADDI	72.34 $\pm$ 1.32	79.22 $\pm$ 1.31	78.27 $\pm$ 1.30
SSI-DDI(Ours)	73.35 $\pm$ 0.70	80.92 $\pm$ 1.62	80.80 $\pm$ 2.25

$\mathcal{D}_{train}$ contains DDI tuples where the drugs are both reserved for training (seen drugs), whereas $\mathcal{D}_{S1}$ contains DDI tuples with drugs not appearing in any of $\mathcal{D}_{train}$ ’s tuples, the remaining of the DDI tuples, i.e. those with one drug appearing in $\mathcal{D}_{S1}$ ’s tuples and the other drug in $\mathcal{D}_{train}$ ’s, are collected in $\mathcal{D}_{S2}$. The model is trained on $\mathcal{D}_{train}$ and tested on both $\mathcal{D}_{S1}$ and $\mathcal{D}_{S2}$. In Table 2, we compare our method to the baseline methods expect GoGNN. The latter is not suitable for this setting because it relies on the topological information of DDIs, which misses for new (unseen) drugs. We kept the same settings of our model as in Section Setup, but changed the learning rate to $lr = 1e - 3 \times 0.96^i$, and set the ratio k of new drugs to $\frac{1}{5}$. From Table 2, which shows the mean and SD of the results of three runs, we can see that our method (competitive with MHCADDI) distinguishes between interacting and non-interacting pair of drugs with good precision, even on unseen drugs. This is attributed to the reformulation of high-level DDIs to the low-level SSIs. Even though drugs are different, but they can still have similar substructures, which we expect to behave the same in any drug where they appear. Here, again we can see that MR-GNN and DeepDDI have very high AUROC (due to the imbalance dataset, and therefore not reliable), even though their AUPRC scores have dropped significantly.

Nevertheless, there is a noticeable drop in performance from the transductive setting to the inductive setting, whereas MHCADDI, with lower performance in transductive setting, was not much affected in inductive setting. In the [Supplementary materials](#), we conduct more analysis that might explain this drop. We also show experiments results based on scaffold split of the drugs, instead of random split as presented here.

As a further analysis

Given that the validation sets ($\mathcal{D}_{S1}$ and $\mathcal{D}_{S2}$) contain drugs on which the model was not trained on, we also experimented with regularization during the drug feature extraction learning phase of our model by incorporating a dropout layer [51] after each GAT layer. As we can see in Figure 3, the performance on $\mathcal{D}_{S1}$ and $\mathcal{D}_{S2}$ is much more stable, at the expense of $\mathcal{D}_{train}$ —there is less overfitting (Figure 3A) with regularization, and there is a little bit of improvement on the AUPRC, especially on $\mathcal{D}_{S1}$ (Figure 3B). These results show us that different training strategies might be needed for different scenarios: the inductive setting might require more regularization during the drug feature extraction phase to avoid overfitting to the drugs on which the model is trained, whereas in a transductive setting, the model is encouraged to learn the drugs in the training set very well, not worrying about overfitting because they are the same appearing in the validation/test sets as well.

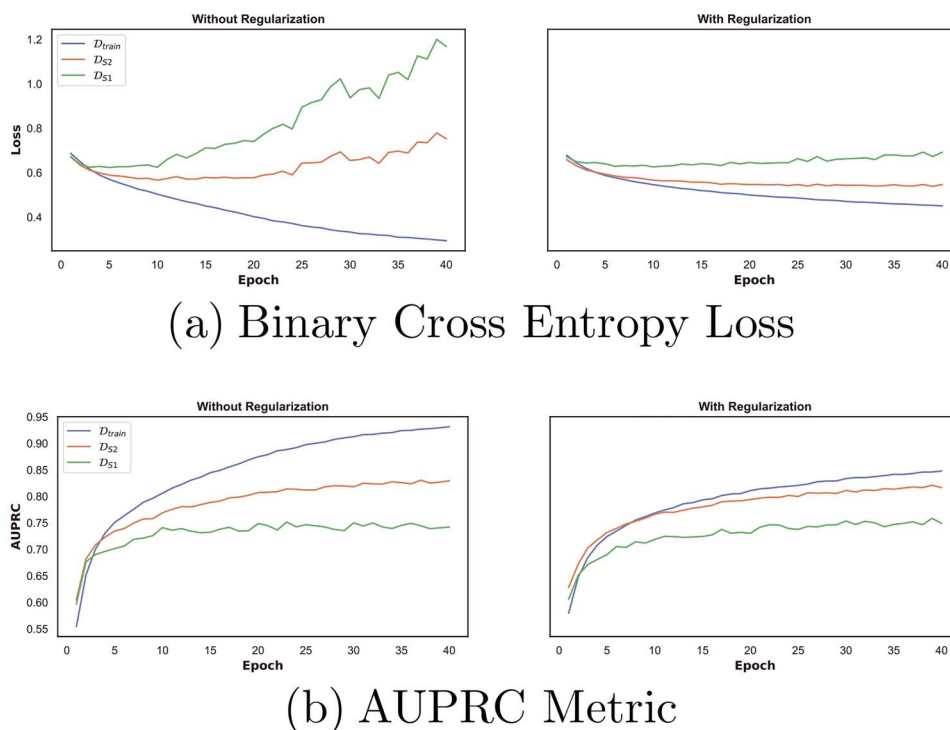


Figure 3. Comparison of loss (A) and AUPRC (B) metrics when regularization is considered in inductive setting during drug feature learning phase.

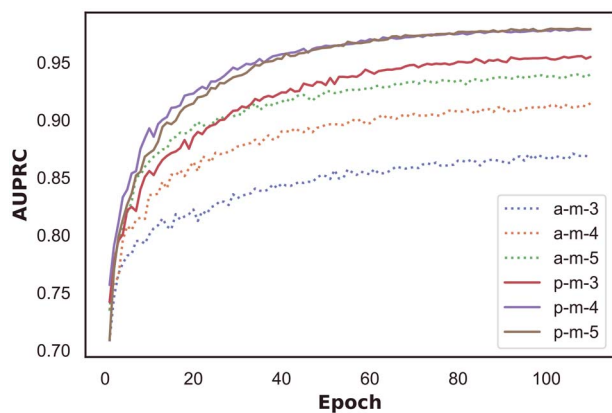


Figure 4. AUPRC for ablation study of co-attention layer: a-m-# refers to an implementation of our method with # GAT layers, but without the co-attention layer, whereas p-m-# refers to the complete (with co-attention) implementation with # GAT layers. We can see co-attention plays a very important role, a shallow model such as p-m-3 outperforms a deep one such as a-m-5.

Ablation study

In order to investigate how the relevance of substructure interactions with co-attention (Section Substructure interaction relevance with co-attention) is a vital component of our method, we conducted experiments by removing the co-attention layer and/or changing the depth of our model by increasing or decreasing the number of GAT layers. The AUPRC results obtained on the validation set in a transductive setting are shown in Figure 4. We can see that the incorporation of co-attention does boost performance: a shallow model (3-GAT layer) with co-attention performs even better than a deep one (5-layer) without co-attention layer.

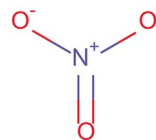


Figure 5. Nitrate functional group.

Case study

In this section, we conduct a case study to investigate substructures that lead to a potential DDI. To that end, we first determine the indices (h, t) of the highest pairwise interaction score from Equation (11):

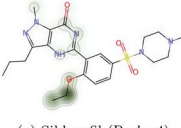
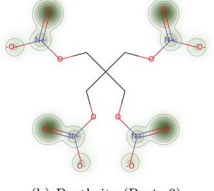
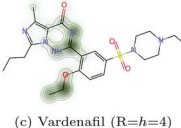
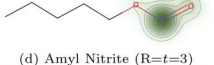
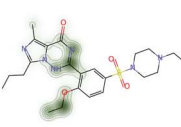
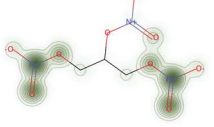
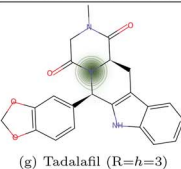
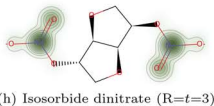
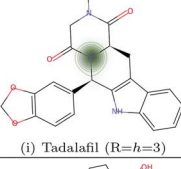
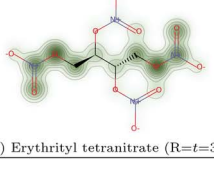
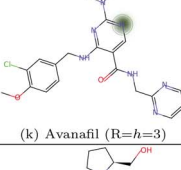
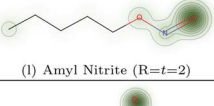
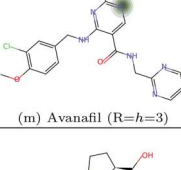
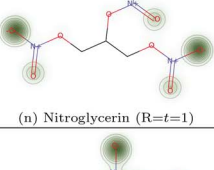
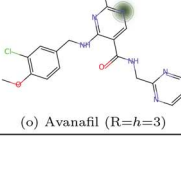
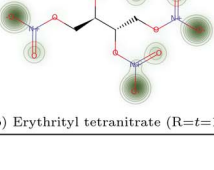
$$(h, t) = \operatorname{argmax}_{i,j} \left(\mathbf{g}_x^{(h)\top} \mathbf{M}_i \mathbf{g}_y^{(j)} \right) \quad (13)$$

$$i = j = 1, \dots, L$$

This can be extended to top k pairwise interaction scores for further analysis, here we use only the highest one (or top 1) to keep the study simple. (h, t) tells us that the interaction between $\mathbf{g}_x^{(h)}$ and $\mathbf{g}_y^{(t)}$ of drugs G_x and G_y , respectively, contributed the most to the final DDI outcome. The substructures of concern are from the h th GAT layer for G_x and t th GAT layer for G_y (see Section Chemical substructure extraction). Furthermore, since a node at the l th GAT layer represents a substructure composed of itself, and its 1- to l -hop neighbors, we can infer that substructures of interest of G_x have a radius of size h , and those of G_y of size t . Finally, from Equation (9) we determine the weight of each substructure.

For a demonstration, we consider the case of phosphodiesterase-5 (PDE5) inhibitors (namely, Sildenafil, Vardenafil,

Table 3. Contributions of substructures to DDI prediction outcomes between some PDE5 inhibitors and organic nitrate-based drugs

PDE5 Inhibitor	DDI Probability	Nitrate Drug
 (a) Sildenafil ($R=h=4$)	$\leftarrow p = 0.9987 \rightarrow$	 (b) Penthrine ($R=t=3$)
 (c) Vardenafil ($R=h=4$)	$\leftarrow p = 0.9968 \rightarrow$	 (d) Amyl Nitrite ($R=t=3$)
 (e) Vardenafil ($R=h=4$)	$\leftarrow p = 0.9867 \rightarrow$	 (f) Nitroglycerin ($R=t=3$)
 (g) Tadalafil ($R=h=3$)	$\leftarrow p = 0.9960 \rightarrow$	 (h) Isosorbide dinitrate ($R=t=3$)
 (i) Tadalafil ($R=h=3$)	$\leftarrow p = 0.9967 \rightarrow$	 (j) Erythrityl tetranitrate ($R=t=3$)
 (k) Avanafil ($R=h=3$)	$\leftarrow p = 0.8849 \rightarrow$	 (l) Amyl Nitrite ($R=t=2$)
 (m) Avanafil ($R=h=3$)	$\leftarrow p = 0.5652 \rightarrow$	 (n) Nitroglycerin ($R=t=1$)
 (o) Avanafil ($R=h=3$)	$\leftarrow p = 0.6808 \rightarrow$	 (p) Erythrityl tetranitrate ($R=t=1$)

Tadalafil, and Avanafil) used for erectile dysfunction [20, 52]. Coadministration of PDE5 inhibitors with organic nitrate (Figure 5) drugs is contraindicated because it can result in synergistic reduction in blood pressure. We want to test if the nitrate functional group has a high contribution to the DDI prediction between a PDE5 inhibitor and a organic nitrate drugs (such

as Penthrine, Amyl Nitrite, Nitroglycerin, Isosorbide dinitrate, Erythrityl tetranitrate). We only keep combinations of PDE5 inhibitors and nitrate drugs that do not appear in the training dataset $\mathcal{D}$.

Table 3 shows the results of this case study. Contributions of substructures are presented as a heat map (contour map with green fill) of the molecular graph. Each row contains a pair of drugs, a PDE5 inhibitor (first column) and a nitrate drug (third column), represented in their molecular graph forms, and their DDI probability (middle column). For each pair of drugs, the indices (h, t) Equation (13) mean that each atom in the molecular graph is the center of a substructure of radius $R = h$ for PDE5 inhibitors, and $R=t$ for nitrate drugs. Therefore, in the heat map, each substructure contribution is shown mainly concentrated around its center. We can see that the first 6 DDIs have very high probabilities of occurring, and the nitrate group (Figure 5) is very much involved in the prediction outcome as the heat map shows. On the other hand, the low probability values of the last two rows can be explained by the fact that the nitrate group was not captured completely. The substructures with high contributions are centered around the oxygens of the nitrate group, and are of radius 1, which is not sufficient to capture the whole functional group. Moreover, contrasting Nitroglycerin in (n) and (f), we can see how the nitrate group involvement differ significantly, similar situation can also be observed with Erythrityl tetranitrate in (j) and (p). More visualizations of substructures relevance to DDIs are presented in the [Supplementary materials](#) where we consider the cases of sulfonylureas and fibrates [53], and those of cephalosporins and Picosulfuric acid⁴.

Conclusion

We propose SSI-DDI, an knowledge-driven deep learning framework for predicting adverse DDIs between a pair of drugs. We demonstrated empirically the power of reformulating the DDI prediction task to a SSI prediction by showing an improvement in performance over previous methods, an improvement both in transductive and inductive (cold-start scenario) settings. Finally, as a case study we showed how the relevance of each SSI during a DDI prediction can be leveraged to gain more insight in the interpretability of the DDI prediction results. One limitation with our method is that performance is affected when the order of drugs of DDIs are changed, even during the training phase. Furthermore, we believe that there is still some noisy information leaked in during the substructure extraction phase which might have affected performance in inductive setting. We intend to investigate it even further in future work for improvement.

Key Points

- Adverse DDI are caused by substructures (function groups) of drug compounds when co-existing in the organism.
- We propose a model based on this knowledge, SSI-DDI, which predicts DDI between two drugs based on interaction between their respective substructures.
- We use an attention mechanism (co-attention) to learn the relevance of each substructure interaction to the final DDI prediction. This relevance index can also be used for the interpretation of the prediction results.

4 <https://go.drugbank.com/drugs/DB09268>

Supplementary Data

Supplementary data are available online at *Briefings in Bioinformatics*.

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